

IT_Help_DESK

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GitHub Repository:

<https://github.com/AbbasMoussawi/IT-Help-Desk.git>

Database Design and System Workflow

1. Database Overview

The IT Help Desk & Ticketing Management System uses a relational database to store users, roles, departments, tickets, comments, notifications, and attachments. The database is designed to support authentication, role-based access, ticket creation, ticket assignment, ticket status tracking, file uploads, and communication between employees and IT support agents.

The database contains the following tables:

User

Department

Role

Category

Priority

Status

Ticket

Comment

Notification

Attachment

2. Database Tables

2.1 Role Table

The Role table stores the different user roles in the system. Roles are used to control what each user can access and do inside the system.

Column Name	Description
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RoleId	Primary key. Unique identifier for each role.
RoleName	Name of the role, such as Admin, Employee, IT Support Agent, or Manager.
CreatedAt	Date and time when the role was created.

Example roles:

Admin
Employee
IT Support Agent
Manager

2.2 Department Table

The Department table stores company departments. Each user can belong to one department, and tickets can also be related to a department.

Column Name	Description
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DepartmentId	Primary key. Unique identifier for each department.
DepartmentName	Name of the department, such as IT, HR, Finance, or Sales.
CreatedAt	Date and time when the department was created.

2.3 User Table

The User table stores all system users. These users can be employees, IT support agents, managers, or admins. Since the system does not include public registration, users are created by the admin.

Column Name Description

UserId	Primary key. Unique identifier for each user.
FullName	Full name of the user.
Email	User email address used for login.
PasswordHash	Encrypted password used for secure authentication.
RoleId	Foreign key connected to the Role table. Defines the user's permissions.
DepartmentId	Foreign key connected to the Department table. Defines the user's department.
IsActive	Shows whether the user account is active or disabled.
CreatedAt	Date and time when the user was created.
UpdatedAt	Date and time when the user information was last updated.

2.4 Category Table

The Category table stores ticket categories. Categories help classify the type of IT issue.

Column Name Description

CategoryId	Primary key. Unique identifier for each category.
CategoryName	Name of the category.
Description	Optional explanation of the category.
CreatedAt	Date and time when the category was created.

Example categories:

Hardware
Software
Network
Email
Access Request
Other

2.5 Priority Table

The Priority table stores ticket urgency levels. It helps IT support decide which tickets should be handled first.

Column Name Description

PriorityId Primary key. Unique identifier for each priority.

PriorityName Name of the priority level.

PriorityLevel Numeric value used for sorting priorities.

CreatedAt Date and time when the priority was created.

Example priorities:

Low

Medium

High

Critical

2.6 Status Table

The Status table stores the possible stages of a ticket.

Column Name Description

StatusId Primary key. Unique identifier for each status.

StatusName Name of the ticket status.

CreatedAt Date and time when the status was created.

Example statuses:

Open

In Progress

Pending

Resolved

Closed

2.7 Ticket Table

The Ticket table is the main table in the system. It stores all support requests created by employees.

Column Name	Description
TicketId	Primary key. Unique identifier for each ticket.
TicketNumber	Unique reference number for the ticket, such as TCK-0001.
Title	Short title of the issue.
Description	Full description of the problem.
CreatedByUserId	Foreign key connected to the User table. Shows who created the ticket.
AssignedToUserId	Foreign key connected to the User table. Shows which IT support agent is assigned to the ticket.
DepartmentId	Foreign key connected to the Department table. Shows the department related to the ticket.
CategoryId	Foreign key connected to the Category table. Shows the type of issue.
PriorityId	Foreign key connected to the Priority table. Shows the urgency of the ticket.
StatusId	Foreign key connected to the Status table. Shows the current ticket stage.
CreatedAt	Date and time when the ticket was created.
UpdatedAt	Date and time when the ticket was last updated.
ResolvedAt	Date and time when the ticket was resolved.
ClosedAt	Date and time when the ticket was closed.

2.8 Comment Table

The Comment table stores comments and replies related to tickets. Employees, IT support agents, and admins can add comments to communicate about a ticket.

Column Name	Description
CommentId	Primary key. Unique identifier for each comment.
TicketId	Foreign key connected to the Ticket table. Shows which ticket the comment belongs to.
UserId	Foreign key connected to the User table. Shows who wrote the comment.
CommentText	The comment or reply text.
IsInternal	Shows whether the comment is internal and only visible to IT/Admin users.
CreatedAt	Date and time when the comment was created.

2.9 Notification Table

The Notification table stores alerts sent to users when something happens in the system, such as a ticket being assigned, updated, commented on, resolved, or closed.

Column Name	Description
NotificationId	Primary key. Unique identifier for each notification.
UserId	Foreign key connected to the User table. Shows who receives the notification.
TicketId	Foreign key connected to the Ticket table. Shows which ticket the notification is related to.
Message	Notification message shown to the user.
IsRead	Shows whether the user has read the notification.
CreatedAt	Date and time when the notification was created.
ReadAt	Date and time when the notification was read.

2.10 Attachment Table

The Attachment table stores files uploaded to tickets, such as screenshots, documents, error logs, or PDFs.

Column Name	Description
AttachmentId	Primary key. Unique identifier for each attachment.
TicketId	Foreign key connected to the Ticket table. Shows which ticket the file belongs to.
UploadedByUserId	Foreign key connected to the User table. Shows who uploaded the file.
FileName	Original name of the uploaded file.
FilePath	Secure file storage path or URL.
FileType	File type, such as PNG, JPG, PDF, DOCX, or LOG.
FileSize	File size in bytes.
CreatedAt	Date and time when the file was uploaded.

3. Table Relationships

The database relationships define how the tables are connected to each other.

Relationship	Description
Role 1 → many User	One role can be assigned to many users.
Department 1 → many User	One department can include many users.
Department 1 → many Ticket	One department can be related to many tickets.
User 1 → many Ticket as creator	One user can create many tickets.
User 1 → many Ticket as assigned agent	One IT support agent can be assigned to many tickets.
User 1 → many Comment	One user can write many comments.
User 1 → many Notification	One user can receive many notifications.
User 1 → many Attachment	One user can upload many attachments.
Category 1 → many Ticket	One category can classify many tickets.
Priority 1 → many Ticket	One priority level can be used by many tickets.
Status 1 → many Ticket	One status can be used by many tickets.
Ticket 1 → many Comment	One ticket can have many comments.
Ticket 1 → many Notification	One ticket can generate many notifications.
Ticket 1 → many Attachment	One ticket can have many uploaded files.

4. System Workflows

4.1 Login Workflow

This workflow explains how users access the system.

User opens the login page



User enters email and password



System checks the email and password



System verifies the user role



System allows access based on the user role



User is redirected to the correct dashboard

Roles may include:

Admin

Employee

IT Support Agent

Manager

4.2 Admin Creates User Workflow

Since the system does not include public registration, users are created by the admin.

Admin logs in



Admin opens Manage Users page



Admin enters user information



Admin selects role and department



System creates the user account



User can log in using the created account

4.3 Employee Creates Ticket Workflow

This workflow explains how an employee submits an IT support request.

Employee logs in



Employee opens Create Ticket page



Employee enters ticket title and description



Employee selects category and priority



Employee uploads optional attachment



Employee submits the ticket



System creates the ticket with status Open



System sends notification to admin or IT support

4.4 Ticket Assignment Workflow

This workflow explains how an admin or IT support agent handles ticket assignment.

Admin or IT support agent logs in



User opens Ticket List page



User views open tickets



Admin assigns ticket to an IT support agent



System updates the AssignedToUserId field



System sends notification to the assigned agent



Assigned agent starts working on the ticket

4.5 Ticket Status Update Workflow

This workflow explains how tickets move through different statuses.

IT support agent opens assigned ticket



Agent reviews the ticket details



Agent changes status from Open to In Progress



Agent adds comments or internal notes



If more information is needed, status changes to Pending



When the issue is solved, status changes to Resolved



After confirmation, status changes to Closed



System updates ticket dates and sends notifications

4.6 Comment Workflow

This workflow explains how users communicate inside a ticket.

User opens Ticket Details page



User writes a comment



System saves the comment under the ticket



System sends notification to related users



Other users can view and reply to the comment

Internal comments can be used by IT support agents or admins for private notes.

4.7 Attachment Workflow

This workflow explains how files are uploaded to tickets.

User opens Create Ticket or Ticket Details page



User selects a file to upload



System checks file type and file size



System stores the file securely



System saves file information in the Attachment table



Users with permission can download or view the attachment

4.8 Notification Workflow

This workflow explains how notifications are created.

Ticket is created, assigned, updated, commented on, resolved, or closed



System creates a notification record



Notification is linked to the related user and ticket



User sees notification in the notification center



User opens the notification



System marks it as read

5. Summary

The database design successfully supports the core functionalities of the IT Help Desk & Ticketing Management System.

It manages users, roles, departments, tickets, comments, notifications, and attachments while maintaining secure and organized relationships between all entities.

The ERD diagram below shows the relationships between the database tables.

5.1 Entity Relationship Diagram (ERD)

The following figures present the Entity Relationship Diagram (ERD) of the IT Help Desk & Ticketing Management System database.

Figure 1 shows the complete ERD, while Figures 2 and 3 provide zoomed views of specific sections for better readability.

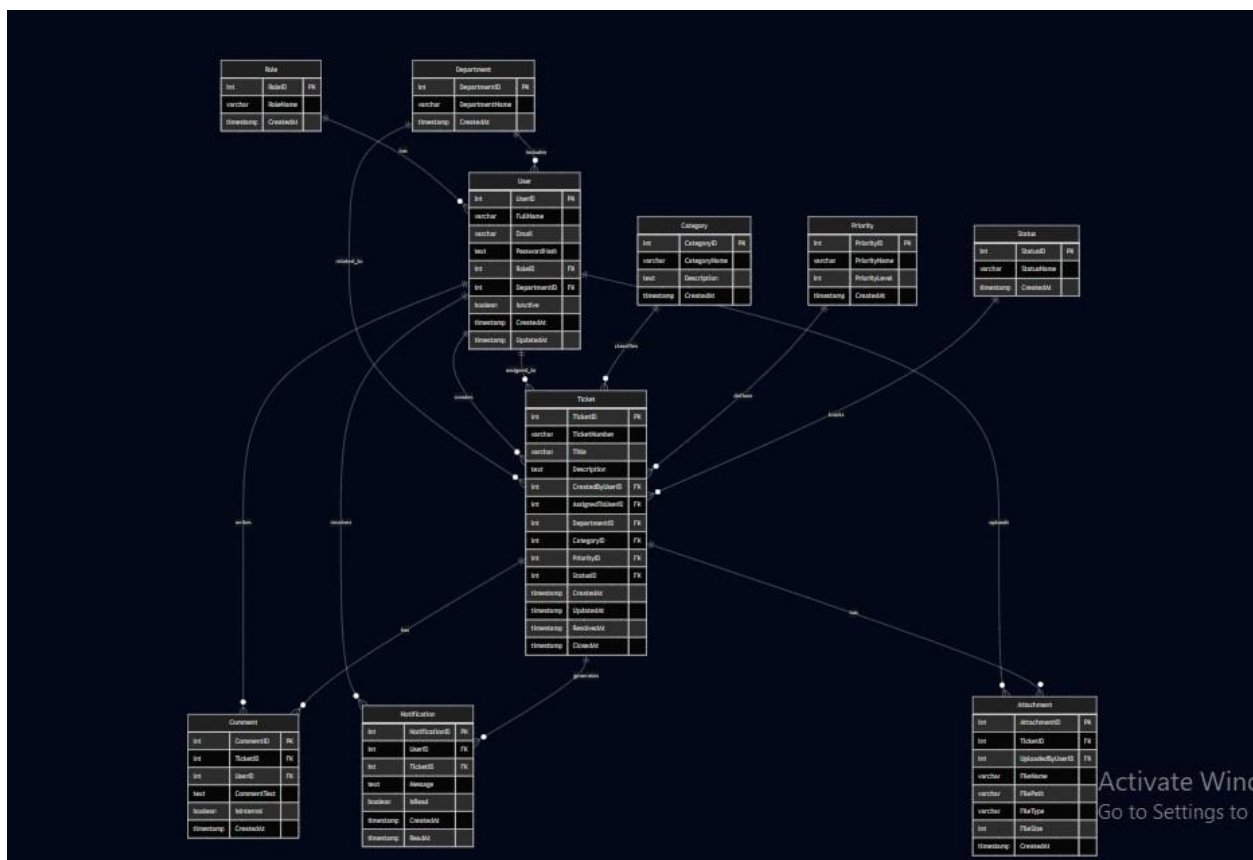


Figure 1: Complete ER Diagram of the Help Desk System Database

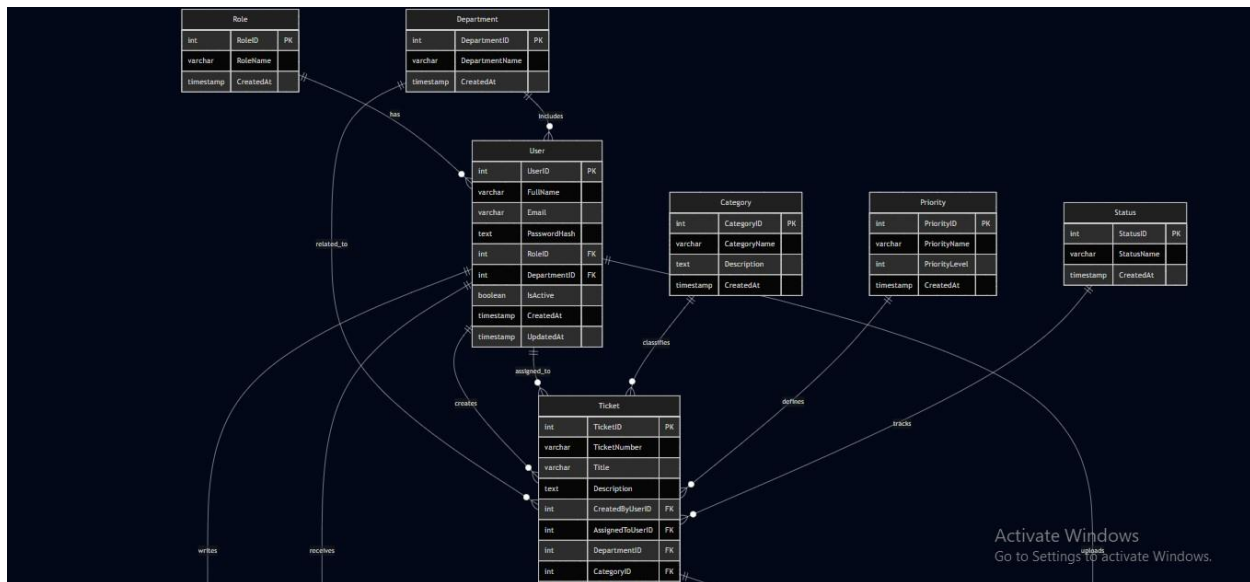


Figure 2: Zoomed View of User, Role, Department, Category, Priority, Status and Ticket Relationship.

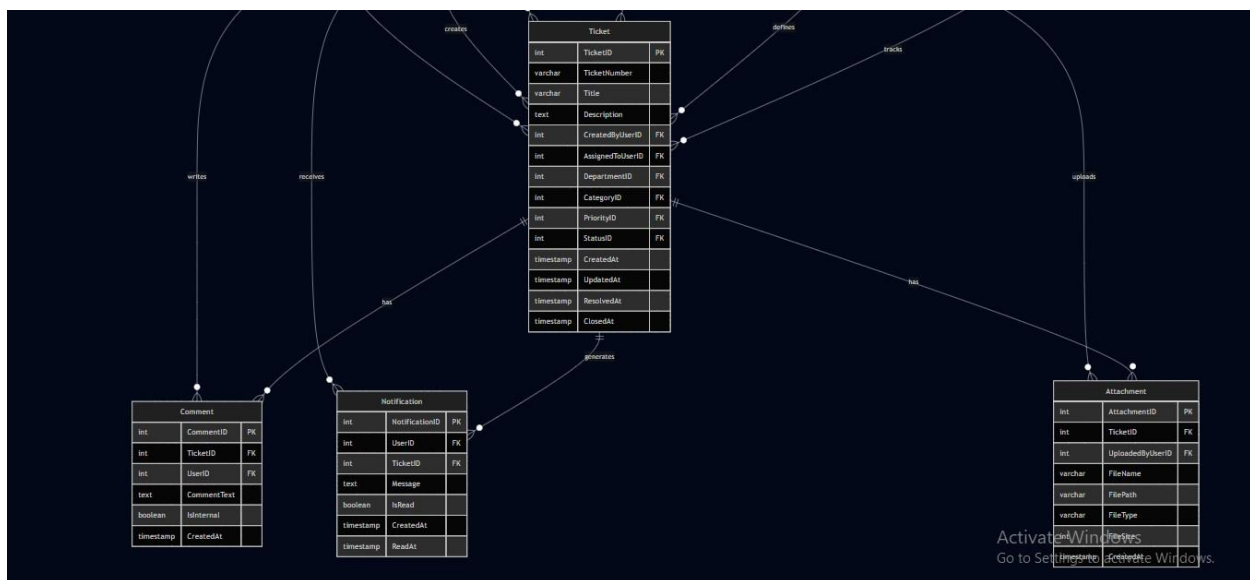


Figure 3: Zoomed View of Ticket, Comment, Notification, and Attachment Relationships.

The ERD demonstrates the overall database structure of the IT Help Desk & Ticketing Management System and illustrates how the different entities interact with each other through relational connections.

These relationships ensure organized data management, secure access control, and efficient ticket handling throughout the system.

6. Authentication and Ticket Management Interfaces

The following interface screens demonstrate the authentication process and ticket management functionalities of the IT Help Desk & Ticketing Management System.

These interfaces were designed to provide a simple, organized, and user-friendly experience for both employees and IT support staff.

6.1 Login Page

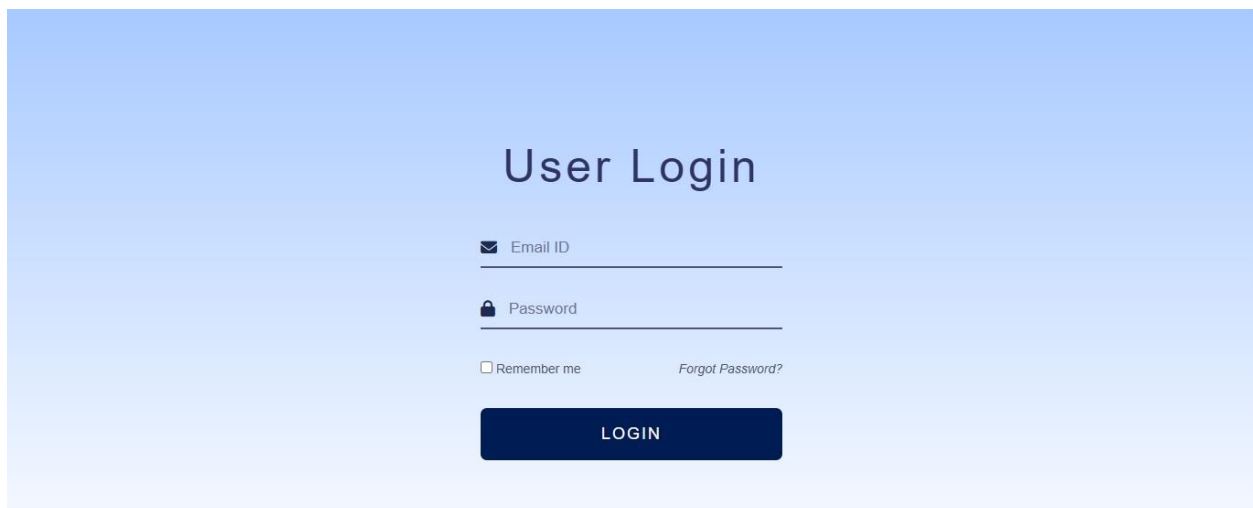


Figure 4: Login Page

The login page allows registered users to securely access the system using their email address and password.

The interface was designed with a clean and simple layout to improve usability and provide quick access to the system functionalities based on the user's assigned role.

6.2 Forgot Password Page

Figure 5: Forgot Password Page

This page enables users to recover access to their accounts in case they forget their password.

Users are required to enter their registered email address in order to receive a verification code or password reset instructions securely.

6.3 Verify Code Page

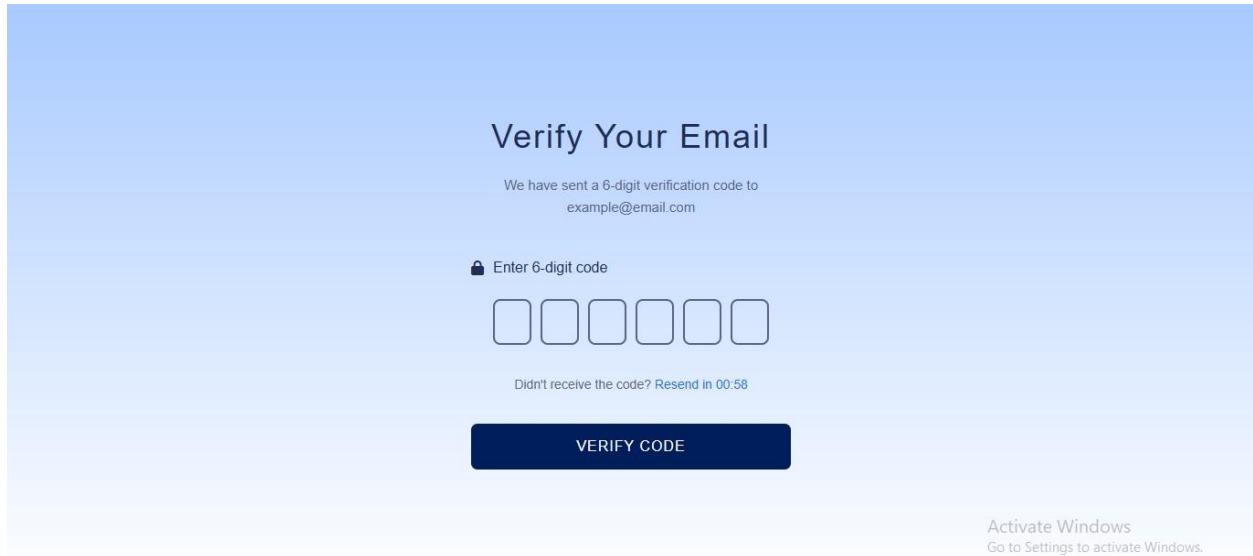
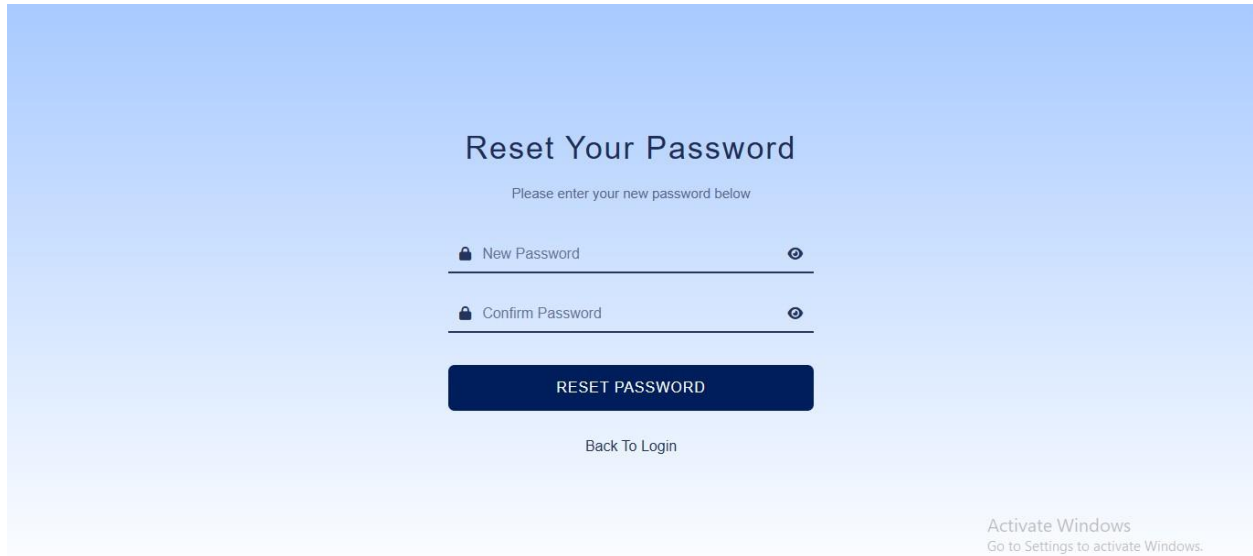
The image shows a web page titled "Verify Your Email" with a light blue gradient background. The title is in a dark blue font. Below the title, a message states: "We have sent a 6-digit verification code to example@email.com". Underneath this, there is a lock icon followed by the text "Enter 6-digit code". Below the text are six empty square input boxes for the code. A link "Resend in 00:58" is positioned below the input boxes. At the bottom center is a dark blue button with the text "VERIFY CODE" in white. In the bottom right corner, there is a small "Activate Windows" watermark with the text "Go to Settings to activate Windows."

Figure 6: Verify Code Page

The verification page is used to confirm the identity of the user during the password recovery process.

Users must enter the verification code sent to their registered email address before being allowed to reset their password.

6.4 Reset Password Page



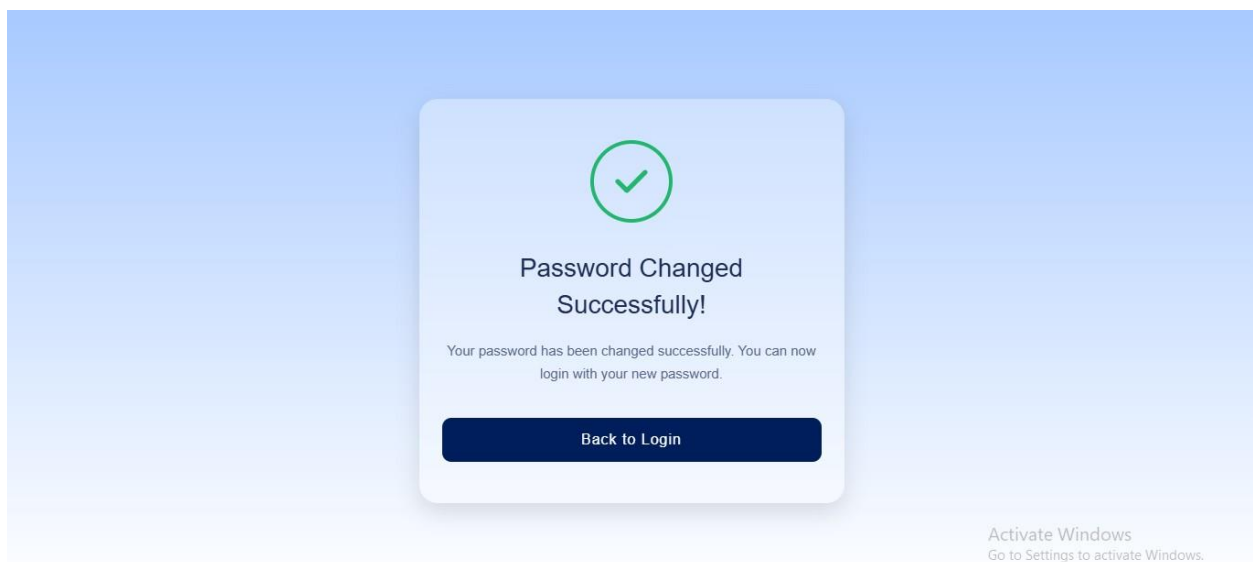
The image shows a 'Reset Your Password' form on a light blue background. The form is centered and contains the following elements: a title 'Reset Your Password', a subtitle 'Please enter your new password below', two input fields labeled 'New Password' and 'Confirm Password' (each with a lock icon and a toggle eye icon), a dark blue 'RESET PASSWORD' button, and a 'Back To Login' link. In the bottom right corner, there is a 'Activate Windows' watermark with the text 'Go to Settings to activate Windows.'

Figure 7: Reset Password Page

This page allows users to create a new secure password after successful verification.

The interface ensures that users can safely update their account credentials and regain secure access to the system.

6.5 Password Reset Success Page



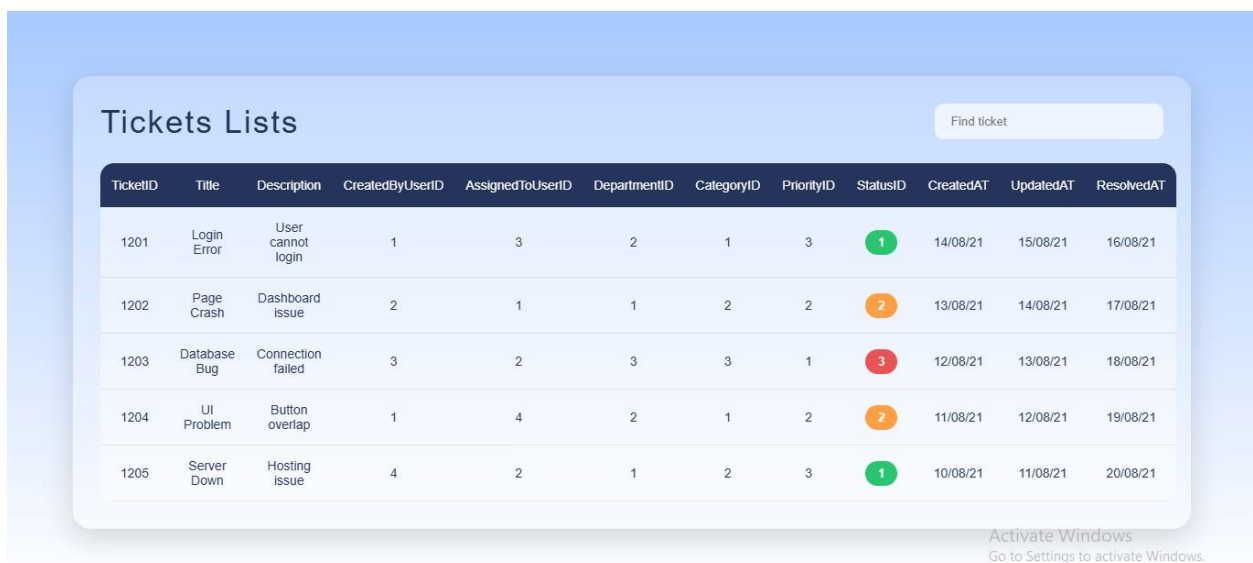
The image shows a 'Password Reset Success' message on a light blue background. The message is centered in a white box with a green checkmark icon. The text reads: 'Password Changed Successfully!', 'Your password has been changed successfully. You can now login with your new password.', and a dark blue 'Back to Login' button. In the bottom right corner, there is a 'Activate Windows' watermark with the text 'Go to Settings to activate Windows.'

Figure 8: Password Reset Success Page

This page confirms that the password reset process has been completed successfully.

It informs the user that their account password has been updated and that they can now log in securely using their new credentials.

6.6 Tickets List Page



The screenshot shows a web interface for 'Tickets Lists'. It features a search bar labeled 'Find ticket' and a table with 12 columns: TicketID, Title, Description, CreatedByUserID, AssignedToUserID, DepartmentID, CategoryID, PriorityID, StatusID, CreatedAT, UpdatedAT, and ResolvedAT. The table contains five rows of ticket data. The StatusID column uses colored circles with numbers to indicate status: green for 1, orange for 2, and red for 3. At the bottom right, there is a watermark for 'Activate Windows'.

TicketID	Title	Description	CreatedByUserID	AssignedToUserID	DepartmentID	CategoryID	PriorityID	StatusID	CreatedAT	UpdatedAT	ResolvedAT
1201	Login Error	User cannot login	1	3	2	1	3	1	14/08/21	15/08/21	16/08/21
1202	Page Crash	Dashboard issue	2	1	1	2	2	2	13/08/21	14/08/21	17/08/21
1203	Database Bug	Connection failed	3	2	3	3	1	3	12/08/21	13/08/21	18/08/21
1204	UI Problem	Button overlap	1	4	2	1	2	2	11/08/21	12/08/21	19/08/21
1205	Server Down	Hosting issue	4	2	1	2	3	1	10/08/21	11/08/21	20/08/21

Figure 9: Tickets List Page

The Tickets List page displays all submitted support tickets in a structured table format for easier management and monitoring.

The interface provides important information such as ticket identifiers, titles, assigned users, departments, categories, priorities, statuses, and timestamps.

This page helps administrators and IT support staff efficiently track, organize, and manage technical support requests within the system.

6.7 Ticket Details Page

Ticket Details

TCK-0001

Laptop not connecting to Wi-Fi

In Progress

Created on: 24 May 2026

 Created By
Sara Ali

 Assigned To
Ahmad Khaled

 Category
Network

 Priority
High

Description

The employee is unable to connect the laptop to the office Wi-Fi network. The issue started this morning after restarting the device.

Update Status

In Progress

SAVE CHANGES

Attachments

 screenshot-error.png

Download

 wifi-log.txt

Download

Comments

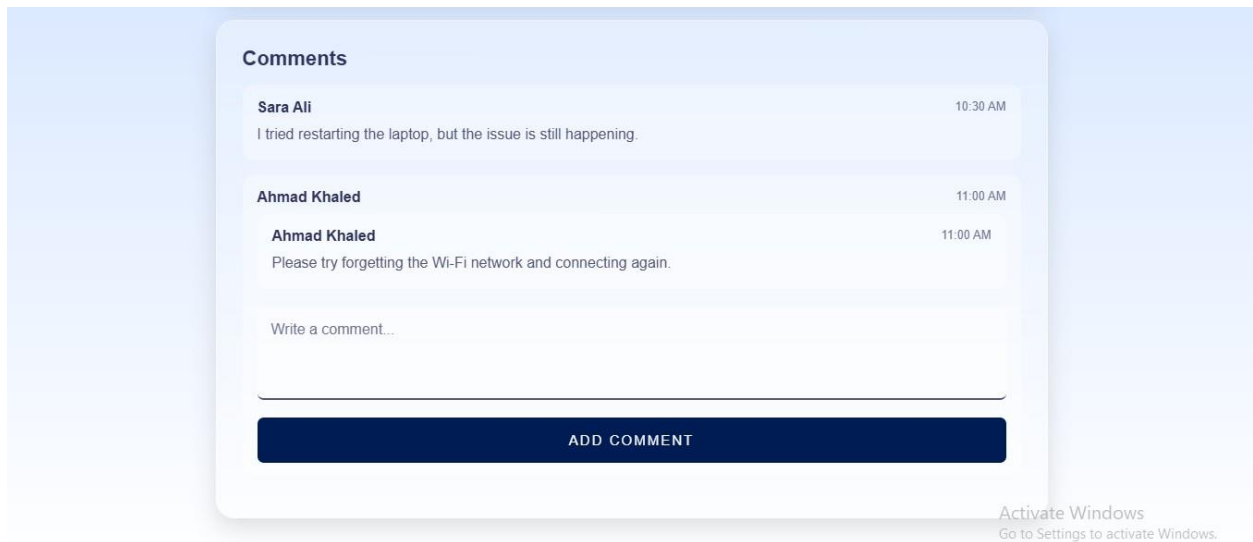
Sara Ali10:30 AM

I tried restarting the laptop, but the issue is still happening.

Ahmad Khaled11:00 AM

Ahmad Khaled11:00 AM

Please try forgetting the Wi-Fi network and connecting again.



Figures 10, 11, 12: Ticket Details Page

The Ticket Details page provides a comprehensive overview of a specific support ticket within the system.

It displays detailed information including the ticket title, description, category, priority level, assigned IT support employee, current status, and creation details.

Additionally, the page allows users and support agents to manage ticket progress, upload attachments, review communication history through comments, and provide updates until the issue is fully resolved.

Overall, the presented user interface screens demonstrate the main functionalities of the IT Help Desk & Ticketing Management System.

The interfaces were designed to provide a clear, organized, and user-friendly experience that simplifies authentication, ticket management, communication, and issue tracking processes for both employees and IT support staff.